AMRUTHA SM

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PROFESSIONAL SUMMARY

Data-driven Computer Science undergraduate (CGPA 9.59) with hands-on experience in machine learning, data analytics, and predictive modeling. Accomplished end-to-end model development — from data cleaning and exploratory analysis to deployment — achieving up to 92% accuracy in classification and regression tasks. Skilled in Python, Pandas, Scikit-learn, and visualization tools to extract insights and guide decision-making. Passionate about leveraging data to build scalable, intelligent solutions that optimize performance and enable business value.

EDUCATION

B.Tech – Computer Science and Engineering
SRM Institute of Science and Technology, Ramapuram | 2027 | CGPA: 9.59/10
Class XII CBSE | 2023 | 87.8%
Class X CBSE | 2021 | 82.4%

PROJECTS

DocIQ — AI Document Intelligence System *(Node.js, React, PostgreSQL, Groq AI)*

- Built a full-stack AI-powered document intelligence platform that extracts structured information from PDFs, images, and text files using Groq's LLaMA model
- Developed REST API using Node.js and Express with Prisma ORM connected to PostgreSQL for storing documents, extractions, and chat sessions
- Implemented real-time document Q&A chat feature for users to ask questions about uploaded documents using AI

Face Recognition Attendance System (Python, OpenCV, Firebase)

- Accomplished automated attendance tracking by implementing real-time facial recognition algorithms using OpenCV and Haar cascades, which reduced manual entry time by 90%.
- Integrated Firebase for cloud-based storage and synchronization, improving data retrieval speed by 60% and ensuring reliable record management.

Spam Mail Detection System (Python, Scikit-learn, Streamlit)

- Built a machine learning classifier using TF-IDF vectorization and Naïve Bayes, achieving 97% precision and 95% recall on test data.
- Performed feature engineering and text preprocessing (tokenization, stopword removal, stemming) to enhance model efficiency by 20%.

Crop Yield Prediction Model (Python, Flask, Scikit-learn)

- Designed a regression-based ML model using historical rainfall, soil, and temperature data.
- Implemented Flask web deployment with REST API integration, enabling seamless input of user data and real-time output visualization.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++
Web: HTML, CSS, JavaScript, React, Flask, Streamlit
Machine Learning: Supervised Learning, Classification, Regression, NLP (TF-IDF, tokenization), Feature Engineering, EDA
Data & Analytics: Scikit-learn, Pandas, NumPy, Predictive modeling, Data visualization, Insight generation
Databases: MySQL, Firebase, PostgreSQL
Tools: Git, GitHub, VS Code, Figma, Jupyter Notebook
Core: Data Structures & Algorithms, OOPS, DBMS

CERTIFICATIONS

Java Programming (Udemy) | Python 101 for Data Science (Udemy) | Database Management Systems (NPTEL) |
IBM Data Fundamentals (IBM)